Business Impact Assessment: AI Travel Planning Assistant

How an AI travel planner can create value for travel companies and their customers.

1. Problem and Opportunity

Travellers usually switch between flight, hotel, and attraction websites, compare inconsistent information, and build itineraries by hand. Travel companies face high support costs for questions such as 'what should I do in Goa for 3 days on a budget?'. An AI travel planner can answer these questions in one place, with a structured plan and a budget breakdown.

2. Target Users and Companies

Target	Need	How the product helps
Individual travellers	Quick, budget-friendly plans	Single form gives flights, hotels, p
Online travel platforms (MakeMyTrip, ClearTrip, Ixigo)	Self-service planning, lower support load	Embedded planner that answers
Travel agencies	Faster itinerary drafts for clients	Day-wise itinerary and budget in s
Hotels and resorts	Better visibility and bookings	Hotel recommendations filtered b

3. Value for Customers

- Time saving: one request replaces many website searches. A plan that normally takes 1 to 2 hours can be drafted in about a minute.
- Budget clarity: the plan shows flight, hotel, and local costs, and compares the total with the user's budget.
- Better decisions: the agent explains why each flight and hotel was selected, and flags over-budget totals.
- Free local browsing: hotels, places, and itinerary tabs work without any AI cost, so users can explore freely.
- Saved plans: users keep favorites and trip plans, and can export a JSON file for their own tools.

4. Value for Businesses

Benefit	Description	Example measure
Lower support cost	Common trip questions answered by the assistant and planner	Share of support tickets on itinerary an
Higher engagement	Users spend more time on the platform exploring plans	Session length and plan generations pe
Better conversion	Planner links users to hotels and flights they liked	Click-through from plan to booking pag
Faster sales cycle	Agencies produce draft itineraries quickly	Time to first itinerary draft
Data for product decisions	Logs show which destinations and preferences are popular	Most requested cities and interests

5. Cost and Operating Model

The current prototype runs on free tiers: Groq free tier for the LLM, Open-Meteo for weather, and Render free hosting. For a production launch, the main recurring cost is LLM usage. The local tabs (hotels, places, itinerary) have no LLM cost, so most browsing can stay free, and only plan generation and chat use the model.

Cost item	Prototype	Production estimate
LLM (Groq or similar)	Free tier (daily token limit)	Pay per token, depends on plan requests per day
Hosting	Render free plan	Paid web service for always-on use
Weather API	Open-Meteo (free, no key)	Free for non-commercial use, check terms
Data	Sample JSON datasets	Partner APIs or data licences

6. Risks and Mitigation

- Inaccurate data: sample datasets can show wrong prices or schedules. Mitigation: show a clear note on each result, and connect real data sources before launch.

- LLM cost overrun: heavy use can exceed the budget. Mitigation: local tabs for browsing, token-limited tool outputs, and rate limits per user.

- Over-reliance on AI answers: users may trust the plan without checking. Mitigation: disclaimers and a link to official booking pages.

- Privacy: trip details can be sensitive. Mitigation: data stored in the browser only in this prototype, and no personal data sent to the LLM beyond the trip query.

- Service availability: the free hosting plan sleeps when idle. Mitigation: move to a paid plan for production use.

7. Success Metrics

Metric	Target (pilot)	How to measure
Plan completion rate	Above 70% of started plans	Plans generated divided by form submissions
Time to first plan	Under 60 seconds (LLM plans)	Measured from submit to result
Plan accuracy	Above 90% of checked fields correct	Agent performance report test results
Saved trips per user	At least 1 per returning user	Saved trips in storage analytics
Support ticket reduction	10% to 20% for planning questions	Compare before and after launch

8. Recommended Next Steps

- Run a pilot with 20 to 30 real users and collect feedback on plan quality.

- Connect a real hotel and flight data source through a partner API.

- Add user accounts so saved trips sync across devices.

- Measure LLM cost per plan and set a monthly budget.

- Prepare a short demo video for portfolio and investor discussions.